

DATA ANALYTICS MINI PROJECT

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Title : Amazon Products Sales - Data Analysis

Project Domain : Sales & E-Commerce

Dataset Year : 2025

Submission Date : 07-01-2026

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Raw Dataset Link : <https://drive.google.com/file/d/1uMohEZpJ0T3aQ7bIJcFIHeSEPA-QfPOD/view?usp=sharing>

Cleaned Dataset : <https://docs.google.com/spreadsheets/d/1FrLcl-rZvKZgRnial3w95ccXayvFhOQz/edit?usp=sharing&oid=104138233421140273365&rtpof=true&sd=true>

EXCEL DATA CLEANING TRANSFORMING REPORT

1. Dataset Overview

- ★ The dataset was sourced from Kaggle and contains information on over 42,000 Amazon products, including pricing, ratings, reviews, demand indicators, and promotional details.
- ★ The raw dataset contained inconsistent formats, text-based values, and missing data, which required cleaning before analysis.

2. Dataset Structure

Total Columns : 16
String/Text : 12
Decimal/Numeric : 3
Date/Time : 1

3. Tools Used

Microsoft Excel – data cleaning, transformation, and creation of derived columns
Excel formulas – IF, TRIM, SEARCH, VALUE for standardization and categorization
Power BI – data loading, calculated measures, KPI's, slicers, and dashboard visualization

4. Data Cleaning & Preparation

- ★ Text-based rating values such as “4.6 out of 5 stars” were converted into numeric format using a Rating_Clean column.
- ★ Purchase indicators from the bought_in_last_month column were standardized into a Demand_Level column.
- ★ Product prices were categorized into Budget, Mid-Range, and Premium groups.
- ★ Review counts were classified into Few Reviews, Moderate Reviews, and Highly Reviewed, with missing values labeled as No Review.
- ★ Coupon information was standardized into No Coupon, Percentage Coupon, and Amount Coupon categories.
- ★ Sponsorship details were cleaned and mapped into a Sponsored_Status column.

5. Key Derived Columns

- ★ Rating_Clean
- ★ Demand_Level
- ★ Price_Range
- ★ Review_Level
- ★ Coupon_Type
- ★ Sponsored_Status

6. Purpose of Data Cleaning

- ★ The cleaning process ensured consistency, removed ambiguity, and prepared the dataset for accurate analysis.
- ★ The cleaned data was structured to support meaningful insights and interactive dashboard creation in Power BI.

POWER BI VISUALIZATION, INSIGHTS & DASHBOARD REPORT

1. Analysis Overview

- * This analysis was performed using the cleaned Amazon products dataset to understand product performance, customer behavior, pricing patterns, and promotional impact using Power BI visualizations.

2. Descriptive Analysis

- * Budget-priced products constitute the largest portion of the dataset.
- * Most products fall under the Medium demand category, indicating steady customer interest.
- * Highly reviewed products dominate the dataset, reflecting strong customer engagement.

3. Diagnostic Analysis

- * Affordable pricing contributes to higher product demand.
- * Products with higher ratings and review volumes attract more customer attention.
- * Organic listings dominate due to limited reliance on paid sponsorships.

4. Predictive Analysis

- * Products with strong ratings and reviews are likely to maintain consistent demand.
- * Budget and Mid-Range products are expected to outperform Premium products in sales volume.
- * Selective use of coupons may increase customer engagement.

5. Prescriptive Analysis

- * Prioritize promotions for high-rated and highly reviewed products.
- * Apply targeted coupons to Mid-Range products to improve performance.
- * Optimize sponsorship usage by focusing on products with strong organic reach.

6. KPI Summary

- * Approximately 42.68K products were analyzed.
- * The average product rating is 4.39, indicating high customer satisfaction.
- * The average discounted price is 154.07, showing a predominantly affordable pricing strategy.

7. Dashboard Visuals

- * KPI cards summarize key performance metrics.
- * Bar and column charts represent demand and price distribution.
- * Pie and donut charts highlight sponsorship and coupon usage.
- * Scatter, tree map, and line charts provide deeper insight into pricing, ratings, and review behavior.
- * A gauge visual compares average rating against a benchmark target.

8. Overall Review

- * The dashboard provides a clear and structured view of Amazon product performance by combining pricing, demand, reviews, and promotional factors. The use of multiple visuals, KPIs, slicers, and measures enables interactive exploration and supports meaningful insight generation for business understanding.

EXCEL DATA CLEANING :

- 1) The rating column contained text values such as “4.6 out of 5 stars,” which could not be used directly for numerical analysis. A new column, Rating_Clean, was created using an Excel formula to extract only the numeric rating values. This enabled the rating data to be used accurately for calculations and further analysis.

	B	C	D
	rating	Rating_Clean	number_of_review
ivalier M	4.6 out of 5 stars	=Left(82,3)	
e, 240W	4.3 out of 5 stars		
ing Case	4.6 out of 5 stars		
:arbuds,	4.6 out of 5 stars		
ck of an	4.8 out of 5 stars		
CE Color	4.6 out of 5 stars		
uds, Blu	4.5 out of 5 stars		
nd find Y	4.6 out of 5 stars		
overs all	4.0 out of 5 stars		
11-inch	4.7 out of 5 stars		
caline H	4.7 out of 5 stars		8,6
ch Lapt	4.8 out of 5 stars		
eless Ea	4.2 out of 5 stars		
ternal F	4.4 out of 5 stars		1
i Strap,	4.7 out of 5 stars		
cientific	4.7 out of 5 stars		4

	B	C	D
	rating	Rating_Clean	number_of_review
aller M	4.6 out of 5 stars	4.6	
240W	4.3 out of 5 stars	4.3	
g Case	4.6 out of 5 stars	4.6	
rbuds,	4.6 out of 5 stars	4.6	
k of an	4.8 out of 5 stars	4.8	
Color	4.6 out of 5 stars	4.6	
ds, Blu	4.5 out of 5 stars	4.5	
find Y	4.6 out of 5 stars	4.6	
ers all	4.0 out of 5 stars	4.0	
1-inch	4.7 out of 5 stars	4.7	
line H	4.7 out of 5 stars	4.7	
h Lapt	4.8 out of 5 stars	4.8	
less Ea	4.2 out of 5 stars	4.2	
ernal F	4.4 out of 5 stars	4.4	
trap,	4.7 out of 5 stars	4.7	
entific	4.7 out of 5 stars	4.7	
py Pri	4.8 out of 5 stars	4.8	
gh-Per	4.7 out of 5 stars	4.7	
aminat	4.8 out of 5 stars	4.8	
HC	4.6 out of 5 stars	4.6	
Powe	4.8 out of 5 stars	4.8	
40mm	4.6 out of 5 stars	4.6	
Pencil	4.8 out of 5 stars	4.8	
h USB	4.6 out of 5 stars	4.6	
4 Cour	4.7 out of 5 stars	4.7	
latter	4.8 out of 5 stars	4.8	
is with	4.8 out of 5 stars	4.8	
adgho	4.5 out of 5 stars	4.5	
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rtidge	4.6 out of 5 stars	4.6	
SB Mic	4.5 out of 5 stars	4.5	
SSD U	4.6 out of 5 stars	4.6	
) SSD C	4.4 out of 5 stars	4.4	
IView	4.8 out of 5 stars	4.8	
mov C	4.7 out of 5 stars	4.7	

- 2) The bought_in_last_month column contained text-based purchase indicators and missing values, which could not be analyzed directly.

A new column, Demand_Level, was created using conditional logic, where blank values were classified as Low demand, entries with “K+” were classified as High demand, and remaining values were classified as Medium demand. This approach standardized demand information and supported consistent analysis

	E	F	G	H
	bought_in_last_month	Demand_Level	current/discounted_price	price_on_variant
375	300+ bought in past month	=IF(TRIM(E2)="","Low",IF(RIGHT(TRIM(E2),2)="K+","High","Medium"))		
2,457	6K+ bought in past month		9.99	basic variant price: nan
3,044	2K+ bought in past month		314	basic variant price: nan
5,882	10K+ bought in past month			basic variant price: \$162
8,988	10K+ bought in past month			basic variant price: \$72.
4,522	100K+ bought in past month			basic variant price: \$99.
3,466	10K+ bought in past month			basic variant price: \$88.
8,105	10K+ bought in past month		16.99	basic variant price: \$23.
4,380				
7,308	10K+ bought in past month			basic variant price: \$284
5,598	100K+ bought in past month		14.99	basic variant price: nan
1,914	10K+ bought in past month			basic variant price: \$880
153	10K+ bought in past month		26.99	basic variant price: nan

	E	F	G	H
	bought_in_last_month	Demand_Level	current/discounted_price	price_on_variant
375	300+ bought in past month	Medium		
2,457	6K+ bought in past month	Medium		
3,044	2K+ bought in past month	Medium		
5,882	10K+ bought in past month	Medium		
8,988	10K+ bought in past month	Medium		
4,522	100K+ bought in past month	Medium		
3,466	10K+ bought in past month	Medium		
8,105	10K+ bought in past month	Medium		
4,380		Low		
7,308	10K+ bought in past month	Medium		
5,598	100K+ bought in past month	Medium		
1,914	10K+ bought in past month	Medium		
153	100K+ bought in past month	Medium		
12,076	1K+ bought in past month	Medium		
2,235	500+ bought in past month	Medium		
48,466	100K+ bought in past month	Medium		
1,98,512	80K+ bought in past month	Medium		
6,23,776	100K+ bought in past month	Medium		
88,965	100K+ bought in past month	Medium		
1,909	10K+ bought in past month	Medium		
61,157	100K+ bought in past month	Medium		
12,840	10K+ bought in past month	Medium		
54,244	100K+ bought in past month	Medium		
13,087	10K+ bought in past month	Medium		
84,821	100K+ bought in past month	Medium		
43,002	70K+ bought in past month	Medium		
1,23,404	80K+ bought in past month	Medium		
1,08,526	10K+ bought in past month	Medium		
59,092	50K+ bought in past month	Medium		
1,01,072	50K+ bought in past month	Medium		
171		Low		
276		Low		
358		Low		
36,122	70K+ bought in past month	Medium		
1,23,411	10K+ bought in past month	Medium		
44,109	10K+ bought in past month	Medium		
32,810	40K+ bought in past month	Medium		
1,03,375	10K+ bought in past month	Medium		

- 3) Product prices were classified into Budget, Mid-Range, and Premium categories based on numeric price thresholds.

Missing price values were clearly labeled as Not Available to avoid blanks during analysis and visualization.

This categorization enabled consistent price-based comparison across products.

	G	H	I	J	K
	current/discounted_price	Price_Range	price_on_variant	listed_price	is_best_sell
89.68	=IF(G2="", "Not Available", IF(G2<500, "Budget", IF(G2<=1500, "Mid Range", "Premium"))				
9.99			basic variant price: nan	\$15.99	No Badge
314			basic variant price: nan	\$349.00	No Badge

	G	H	I	J	K
	current/discounted_price	Price_Range	price_on_variant	listed_price	is_best_sell
89.68		Budget	basic variant		
9.99		Budget	basic variant		
314		Budget	basic variant		
		Not Available	basic variant		
		Not Available	basic variant		
		Not Available	basic variant		
		Not Available	basic variant		
16.99		Budget	basic variant		
14.99		Not Available	basic variant		
29.99		Budget	basic variant		
249.99		Budget	basic variant		
69.99		Budget	basic variant		
		Not Available	basic variant		
39.97		Budget	basic variant		
14.49		Budget	basic variant		
16.96		Budget	basic variant		
		Not Available	basic variant		
16.95		Budget	basic variant		
		Not Available	basic variant		
9.89		Budget	basic variant		
		Not Available	basic variant		
		Not Available	basic variant		
15.15		Budget	basic variant		
18.79		Budget	basic variant		
		Not Available	basic variant		
		Not Available	basic variant		
		Not Available	basic variant		
134.39		Budget	basic variant		
42.99		Budget	basic variant		
239.99		Budget	basic variant		
16.99		Budget	basic variant		
11.06		Budget	basic variant		
39.97		Budget	basic variant		
		Not Available	basic variant		

- This classification enabled clearer comparison of products based on review intensity**

C		F ₂	Alignment	
Clean	D	E		
number_of_reviews	Review_Level			
375	Moderate Review	300		
2,457	Highly Reviewed	6K+		
3,044	Highly Reviewed	2K+		
35,882	Highly Reviewed	10K		
28,988	Highly Reviewed	10K		
44,522	Highly Reviewed	10K		
13,466	Highly Reviewed	10K		
38,105	Highly Reviewed	10K		
4,380	Highly Reviewed	1K+		
7,308	Highly Reviewed	10K		
8,65,598	Highly Reviewed	10K		
1,914	Highly Reviewed	10K		
152	Moderate Review	100		
12,076	Highly Reviewed	1K+		
7,235	Highly Reviewed	500		
48,468	Highly Reviewed	100		
1,98,512	Highly Reviewed	80K		
6,25,776	Highly Reviewed	100		
88,965	Highly Reviewed	100		
1,909	Highly Reviewed	10K		
61,157	Highly Reviewed	100		
12,840	Highly Reviewed	10K		
54,244	Highly Reviewed	100		
15,087	Highly Reviewed	10K		
84,821	Highly Reviewed	100		
43,002	Highly Reviewed	70K		
1,25,044	Highly Reviewed	80K		
1,08,926	Highly Reviewed	10K		
59,092	Highly Reviewed	50K		
1,01,072	Highly Reviewed	50K		
171	Moderate Review	100		
276	Moderate Review	100		
358	Moderate Review	100		
36,122	Highly Reviewed	70K		
1,23,411	Highly Reviewed	10K		
44,109	Highly Reviewed	10K		
32,810	Highly Reviewed	40K		
1,03,375	Highly Reviewed	10K		

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| | Save 50% with coupon | Percentage Coupon | |
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POWER BI VISUALIZATION :

KPI Explanation :

This screenshot shows the key values calculated from the cleaned dataset. It displays the total number of products included in the analysis, the overall average customer rating, and the average discounted price, summarizing the dataset size, customer satisfaction, and pricing level.

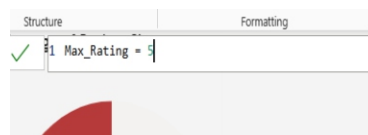
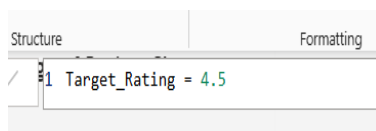


Gauge Measure Creation & Purpose :

To support performance evaluation, calculated measures were created in Power BI.

A target rating measure and a maximum rating measure were defined to enable comparison between actual performance and a benchmark value.

Using these measures, a gauge visual was created to display the average product rating against the target rating. This gauge helps quickly assess overall product quality and indicates how close the current rating is to the expected standard.



Rating Performance Explanation :

This screenshot compares the average product rating with a predefined target value. It shows that the current average rating is close to the target rating, indicating overall strong product quality and positive customer feedback.



Interactive Filters Explanation :

This section allows users to interactively filter the data based on price range, sponsorship status, and demand level.

These filters help focus the analysis on specific product segments and enable dynamic comparison across different categories.

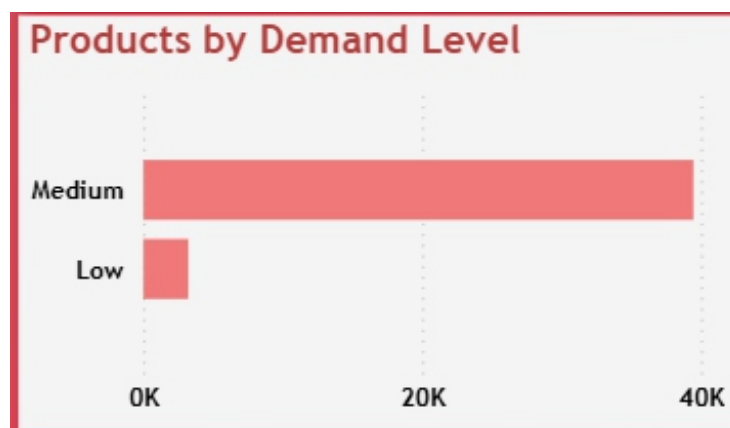
Discount Price_Range
☐ Budaet
☐ Mid Range
☐ Premium

Sponsored_Status
☐ Organic
☐ Sponsored

Demand_Level
☐ Select all
☐ Low
☐ Medium

Products by Demand Level Analysis :

This chart shows the distribution of products based on demand levels derived from recent purchase activity. Most products fall under the Medium demand category, while a smaller portion belongs to the Low demand category, indicating generally steady customer interest across the dataset.



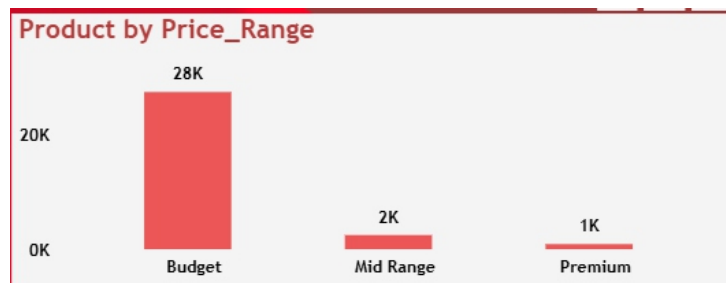
Average Discounted Price by Review Level :

This line chart shows how the average discounted price varies by review level. Products with higher review volumes tend to have lower average prices, suggesting competitive pricing for highly reviewed items.

Products by Price Range :

This chart categorizes products into Budget, Mid-Range, and Premium price segments.

The majority of products are budget-priced, indicating a strong focus on affordable offerings within the dataset.



Coupon_Category Distribution :

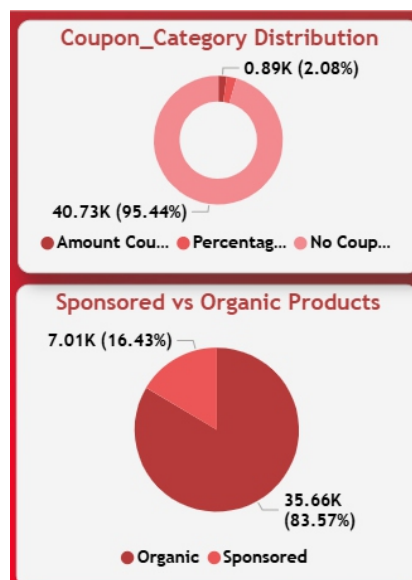
Explanation This chart shows how products are distributed based on coupon availability.

The majority of products fall under the No Coupon category, indicating that most items are sold without promotional discounts. Only a small percentage of products use percentage-based or amount-based coupons, showing that discounts are applied selectively.

Sponsored vs Organic Products :

Explanation This chart compares products based on their sponsorship status. Most products are Organic, meaning they rely on natural visibility rather than paid promotion.

Sponsored products form a smaller portion, indicating limited use of paid advertising across the dataset.



Products by Review Level :

This treemap shows the distribution of products based on review volume. Most products fall under the Highly Reviewed category, indicating strong customer engagement across the dataset.

Products with Moderate Reviews also form a significant portion, while Few Reviews represent a smaller share.

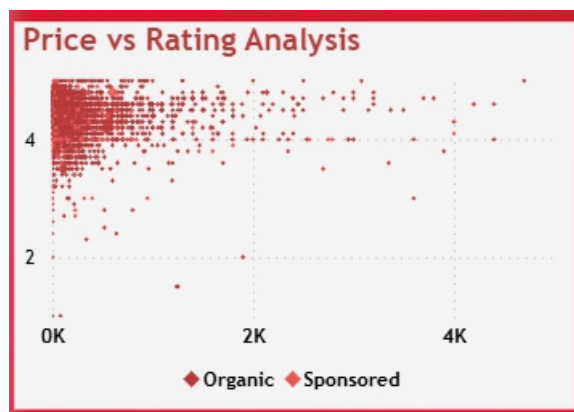
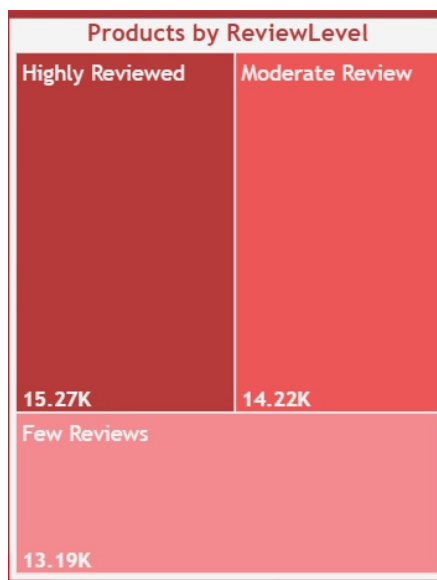
This helps understand overall customer interaction levels with products.

Price vs Rating Analysis :

This scatter chart analyzes the relationship between discounted price and customer rating.

The visualization shows that product ratings remain consistently high across different price levels, suggesting that higher prices do not necessarily lead to lower customer satisfaction.

The color distinction highlights differences between Organic and Sponsored products in terms of pricing and ratings.



Dashboard Explanation :

This dashboard was created using the cleaned Amazon products dataset to analyze product performance and customer behavior.

The analysis focuses on pricing, demand level, customer ratings, review volume, coupons, and sponsorship status.

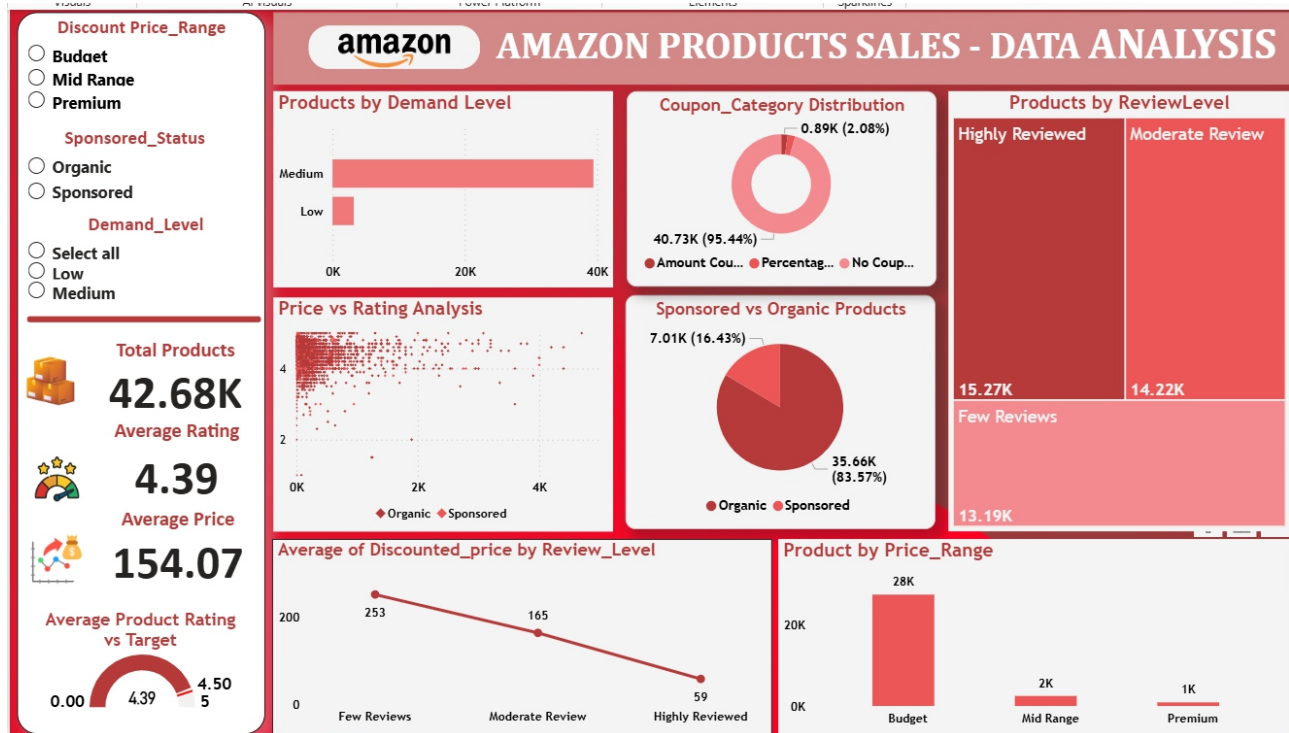
Key KPIs such as total products, average rating, and average price were used to understand the overall dataset summary.

Different visualizations were used to analyze :

- * How products are distributed across price ranges and demand levels
- * The relationship between price and customer ratings
- * Customer engagement through review levels
- * The impact of coupons and sponsorship on products

Interactive slicers were added to filter the data by price range, demand level, and sponsored status, enabling dynamic exploration of insights.

Overall, this dashboard provides a clear and interactive view of Amazon product trends and supports data-driven analysis.



Overall Insights – Amazon Products Analysis :

The dataset is dominated by Budget and Mid-Range products, showing that Amazon's catalog is largely price-sensitive and focused on affordable items.

Most products fall under the Medium Demand category, indicating steady and consistent customer interest across the platform.

Highly Reviewed products form the largest share, highlighting strong customer engagement and active feedback behavior.

The average product rating (4.39) indicates generally high customer satisfaction across different product categories.

The Price vs Rating analysis shows that higher-priced products do not necessarily receive higher ratings, suggesting that price is not the primary driver of customer satisfaction.

Organic products significantly outnumber sponsored products, implying that most items rely on natural visibility rather than paid promotions.

Coupon usage is limited, with No Coupon products forming the majority, indicating selective use of promotional strategies. Products with higher review volumes tend to have lower average prices, suggesting competitive pricing for popular items.

Conclusion :

* This project presents a structured analysis of Amazon product sales data using Excel and Power BI.

The raw dataset was systematically cleaned, transformed, and standardized to address inconsistencies and missing values, ensuring data reliability for analysis.

* Using the cleaned dataset, multiple analytical perspectives were applied to understand product demand, pricing patterns, customer ratings, review behavior, coupon usage, and sponsorship impact.

* The interactive Power BI dashboard effectively visualizes these insights and enables dynamic filtering for deeper exploration.

* Overall, the project demonstrates practical application of data cleaning, analysis, and visualization techniques, and provides meaningful insights that can support data-driven business decision-making.